i. In atoms, the electron revolve around the nucleus in stationary circular paths called orbits or shells or energy levels.
ii. While revolving around the nucleus in an orbit, an electron neither loses nor gains energy.
iii. An electron in a shell can move to a higher or lower energy shell by absorbing or releasing a fixed amount of energy.
iv. The orbits or shells are represented by the letters K,L,M,N,… or the numbers, n= 1,2,3,4,….

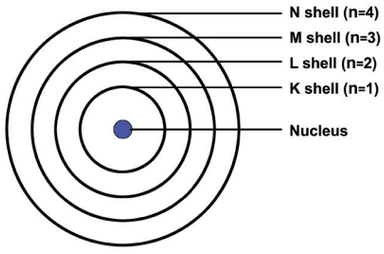


Figure 11.5 Energy levels around the nucleus of an atom: Bohr's model.

The orbit closest to the nucleus is the K shell. It has the least amount of energy and the electrons present in it are called K electrons, and so on with the successive shells and their electrons. These orbits are associated with fixed amount of energy , so Bohr called them as **energy level** or **energy shells**.

11.2.1 Limitations of Bohr's model

One main limitation was that this model was applicable only to hydrogen and hydrogen like ions (example, He^{+}, Li^{2+}, Be^{3+}, and so on). It could not be extended to multi electron nucleus.

11.3 Discovery of Neutrons

In 1932 James Chadwick observed when Beryllium was exposed to alpha particles, particles with about the same mass as protons were emitted.

Beryllium + alpha ray ⟶ carbon + neutron

These emitted particles carried no electrical charges. They were called as neutrons. It is denoted by $_0n^1$. The superscript 1 represents its mass and subscript 0 represents its electric charge.

Properties of Neutrons

1. This particle was not found to be deflected by any magnetic or electric field, proving that it is electrically neutral.
2. Its mass is equal to 1.676×10^{-24} g (**1 amu**).



In 1920 Rutherford predicted the presence of another particle in the nucleus as neutral. James Chadwick, the inventor of neutron was student of Rutherford

11.4 Characteristics of Fundamental particles

The atom is built up of a number of sub-atomic particles. The three sub-atomic particles of great importance in understanding the structure of an atom are electrons, protons and neutrons, the properties of which are given in Table 11.1.

Table 11.1 Properties of sub-atomic particles

Particle	Symbol	Charge (electronic units)	mass (amu)	mass (grams)
Electron	$_{-1}e^{0}$	−1	1/1837	9.1×10^{-28}
Proton	$_1H^1$	+1	1	1.6×10^{-24}
Neutron	$_0n^1$	0	1	1.6×10^{-24}

There are two structural parts of an atom, the nucleus and the empty space in which there are imaginary paths called **orbits**.

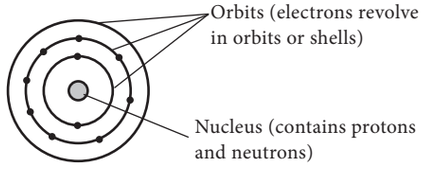


Figure 11.6 Showing structure of an atom.

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